



Islamic University of Technology
Department of Computer Science and Engineering

Lab 2: Sorting

CSE 4404: Algorithms Lab(SWE)
Summer 2024-25

Contents

Task A. The Great Bottle Fill-Off	2
Task Statement	2
Input Format	2
Output Format	2
Task B. Dot Product Minimization	3
Task Statement	3
Input Format	3
Output Format	3
Task C. Plan It Right	4
Task Statement	4
Input Format	4
Output Format	4

Task A. The Great Bottle Fill-Off

Time Limit: 1 second | Memory Limit: 256 MB

You are given a container filled with a limited amount of water. The container can hold at most x liters of water.

You are also given n empty bottles, each with a certain capacity. You want to fill as many bottles as possible using the water in the container, such that each bottle is filled to its full capacity.

Your task is to determine the maximum number of bottles you can completely fill using the water from the container.

Input Format

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains two integers n and x ($1 \leq n \leq 10^4$, $1 \leq x \leq 10^9$) — the number of bottles and the total capacity of the container.
- The second line contains n space-separated integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$) — the amount of water required to completely fill each bottle.

The sum of n over all test cases does not exceed 10^5 .

Output Format

For each test case, print a single integer — the maximum number of bottles you can completely fill.

Sample Input	Sample Output
3	4
5 10	2
2 3 4 5 1	0
4 7	
5 3 4 2	
3 5	
6 7 8	

Task B. Dot Product Minimization

Time Limit: 1 second | Memory Limit: 256 MB

You are given two vectors a and b , each containing n integers. You are allowed to rearrange the elements of each vector in any order.

Your goal is to find such rearrangements that minimize the dot product of the two vectors.

Formally, given two vectors $a = (a_1, a_2, \dots, a_n)$ and $b = (b_1, b_2, \dots, b_n)$, find permutations that minimize:

$$\sum_{i=1}^n a_i \cdot b_i$$

Input Format

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains an integer n ($1 \leq n \leq 10^5$) — the size of the vectors.
- The second line contains n integers $a_1, a_2, \dots, a_n$ ($|a_i| \leq 10^9$).
- The third line contains n integers $b_1, b_2, \dots, b_n$ ($|b_i| \leq 10^9$).

The sum of n over all test cases does not exceed 10^5 .

Output Format

For each test case, print a single integer — the minimum possible dot product.

Sample Input	Sample Output
4	-25
3	60
1 3 -5	-55
-2 4 1	500000000000000
4	
1 2 3 4	
5 6 7 8	
5	
-1 -2 -3 -4 -5	
5 4 3 2 1	
5	
1000000 1000000 1000000 1000000 1000000	
1000000 1000000 1000000 1000000 1000000	

Task C. Plan It Right

Time Limit: 1 second | Memory Limit: 256 MB

You are given a list of n activities, each with a start time and an end time.

A tourist can participate in an activity only if they are not already engaged in another activity at that time.

Your goal is to select the maximum number of non-overlapping activities.

Input Format

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains an integer n ($1 \leq n \leq 10^5$) — the number of activities.
- The second line contains n integers $s_1, s_2, \dots, s_n$ ($1 \leq s_i \leq 10^9$) — start times.
- The third line contains n integers $e_1, e_2, \dots, e_n$ ($s_i \leq e_i \leq 10^9$) — end times.

The sum of n over all test cases does not exceed 10^5 .

Output Format

For each test case, print a single integer — the maximum number of non-overlapping activities.

Sample Input	Sample Output
5	3
4	2
0 5 1 3	2
6 7 2 4	2
3	2
1 4 6	
5 7 10	
6	
3 8 2 1 5 3	
9 9 5 4 9 6	
5	
1 2 3 1 4	
10 10 5 2 5	
3	
1 6 4	
5 10 7	

Marks Distribution

Task	Marks
Task A	40%
Task B	40%
Task C	20%

Practice Problems

Problem	Links
Game	Codeforces: https://codeforces.com/contest/984/problem/A Vjudge: https://vjudge.net/problem/CodeForces-984A
Gravity Flip	Codeforces: https://codeforces.com/contest/405/problem/A Vjudge: https://vjudge.net/problem/CodeForces-405A
Valid Anagram	LeetCode: https://leetcode.com/problems/valid-anagram/
Jagged Swaps	Codeforces: https://codeforces.com/contest/1896/problem/A Vjudge: https://vjudge.net/problem/CodeForces-1896A
Unforgivable Curse (Easy)	Codeforces: https://codeforces.com/contest/1800/problem/E1 Vjudge: https://vjudge.net/problem/CodeForces-1800E1
Unforgivable Curse (Hard)	Codeforces: https://codeforces.com/contest/1800/problem/E2 Vjudge: https://vjudge.net/problem/CodeForces-1800E2